

Integrative Bayesian Inference for Drug Repurposing Using Semantic Priors and Graph Topology

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Abstract

We present a Bayesian framework that integrates semantic priors from biomedical literature with structural signals from a drug–disease graph to estimate posterior probabilities of therapeutic relevance. Semantic priors were derived by classifying PubMed articles using a large language model, distinguishing therapeutic, adverse, and irrelevant associations. These priors were further refined using adverse event reports from FDA data to penalize unsafe signals. Graph-based likelihoods were constructed from interpretable topological features, including graph distance, Katz similarity, and structural likelihood, weighted through supervised learning. Posterior distributions quantified shifts in belief and allowed interpretable ranking of repurposing candidates. To ensure that likelihood contributions were well-founded, we validated the stability of learned feature weights through bootstrap resampling, Gaussian noise injection, and feature ablation. Graph distance consistently emerged as the most predictive signal. Benchmarking against ClinicalTrials.gov showed strong concordance between high-scoring predictions and trial activity, while highlighting overlooked opportunities. This approach enables transparent, data-driven repurposing by fusing literature and clinical network topology.

Keywords: Drug Repurposing, Bayesian Inference, Network Centrality, Machine Learning, Biomedical Literature Mining, Drug–Disease Networks

1 Introduction

Drug repurposing, the repositioning of existing compounds for new therapeutic uses, has emerged as a critical strategy to accelerate drug development, reduce attrition, and contain costs in pharmaceutical innovation. Traditional de novo drug discovery can exceed 12–15 years and cost over \$2 billion, with failure rates nearing 90% in clinical phases. Repurposed drugs, by contrast, benefit from established safety and manufacturing data, which shortens timelines and raises the likelihood of success. Still, finding new and meaningful drug–disease associations is far from straightforward. Biomedical data are vast and fragmented across clinical studies, adverse event records, molecular interactions, and literature. This complexity demands computational models that integrate diverse evidence sources while reducing spurious predictions.

Systematic drug repurposing is gaining momentum as the volume of biomedical data continues to expand. Recent studies have shown the potential of repositioned drugs across a range of diseases, including colchicine in coronary artery disease [Tien et al. \(2021\)](#), multiple therapies for kidney conditions [Panchapakesan and Pollock \(2018\)](#); [Chen et al. \(2024\)](#), and emerging candidates in oncology [Al Khzem et al. \(2024\)](#). Repurposing has also been central in efforts to personalise cardiovascular treatments [Abdelsayed et al. \(2022\)](#), evaluate known compounds in infectious diseases like COVID-19 [Shirazi et al. \(2022\)](#), and identify leads for neurodegenerative disorders [Kakoti et al. \(2022\)](#).

Network-based approaches have proven useful in mapping the therapeutic landscape. Representing drugs and diseases as nodes in a graph makes it possible to detect hidden associations based on structural proximity. Metrics such as Katz similarity and Preferential Attachment help uncover plausible links from existing indication patterns [Askar et al. \(2021\)](#); [Lotfi Shahreza et al. \(2018\)](#). These models have supported the identification of novel mechanisms and enabled the integration of pathway-level and interaction data [Gramatica et al. \(2014\)](#); [Berger and Iyengar \(2009\)](#). Broader frameworks in network medicine reveal shared biological pathways between conditions, offering more avenues for repositioning [Barabási et al. \(2011\)](#). Hybrid approaches combining artificial intelligence with graph analysis have also shown utility in time-sensitive settings, such as COVID-19 [Ahmed et al. \(2022\)](#). Network meta-analysis further extends evidence synthesis by linking indirect comparisons across trials [Tonin et al. \(2017\)](#). However, most network-based tools remain deterministic and lack mechanisms to incorporate prior beliefs or quantify uncertainty.

Advances in natural language processing are now transforming how biomedical literature informs drug discovery. Large language models (LLMs) like GPT-4, Gemini, LLaMA, etc. can identify complex semantic relations in clinical texts and scientific abstracts [Lu et al. \(2025\)](#). Unlike earlier techniques, these models interpret context at scale, improving the extraction of drug–disease and drug–effect associations [Scherbakov et al. \(2025\)](#). In systematic reviews, LLMs have already reduced screening time and improved data extraction performance [Scherbakov et al. \(2025\)](#); [Lu et al. \(2024\)](#). This is especially valuable for repurposing, where relevant evidence is often buried in unstructured text. LLMs thus offer a practical solution for generating semantic priors from literature, allowing evidence-based discovery at a new scale.

Bayesian inference provides a principled way to integrate prior knowledge with

observed data in predictive models. It allows for flexible updates as new evidence emerges and offers a transparent way to express confidence in predictions [Wu and Stephenson \(2024\)](#). This is particularly relevant in repurposing, where data are often noisy or incomplete. Bayesian methods have been adopted in pharmaceutical development for trial design, dose selection, and early efficacy analysis, with support from regulatory agencies [Ionan et al. \(2023\)](#). Their ability to encode expert knowledge or empirical patterns into priors makes them especially suitable for modelling rare diseases and early-stage candidates [Morgan \(2018\)](#). Posterior probabilities from such models not only rank associations but also reflect confidence, guiding more informed decision-making.

Still, unifying semantic insights from literature with structural patterns in networks remains an open challenge. Most models treat these signals separately, limiting their ability to reconcile conflicts or interpret predictions. The lack of transparency in how predictions are formed also reduces trust and clinical usability [Lotfi Shahreza et al. \(2018\)](#).

To address this, we present a Bayesian framework that fuses literature-derived semantic priors with interpretable graph-based features. The model estimates posterior probabilities using weighted likelihoods, balancing textual and structural signals. We also introduce a full evaluation suite focused on assessing the stability and interpretability of feature weights, including feature attribution, ablation studies, and noise resilience tests. This framework supports not only prediction but also transparent inference, enabling scalable identification of high-confidence drug–disease candidates.

2 Method

2.1 Data Collection

Clinical trial data were collected from ClinicalTrials.gov using the public API [of Medicine \(2024\)](#). Trials were filtered to include only drug-based interventions that were part of phase II, III, or IV studies and labeled with a status of recruiting, active not recruiting, enrolling by invitation, completed, withdrawn, or terminated. The search covered ten major therapeutic domains: cancer, diabetes, asthma, Alzheimer’s disease, depression, hypertension, HIV or AIDS, rheumatoid arthritis, epilepsy, and obesity. This yielded 9808 unique trials, each identified by its NCT ID.

From each record, we extracted structured fields including the listed medical conditions, intervention names, study phase, and recruitment status. To ensure terminological consistency, all condition and drug entries were normalised using the 2025 Medical Subject Headings (MeSH) database [of Medicine \(2025\)](#). Matching relied on a combination of exact and fuzzy mapping against MeSH descriptors, with conditions identified through codes beginning with “C” and drugs through codes starting with “D”.

Because a single drug can appear in trials for multiple conditions, and a single condition may be evaluated across many studies, each condition–drug–trial instance was treated separately. This approach preserved the diversity of clinical contexts and allowed finer granularity in downstream analysis. The processed dataset comprised 10178 matched rows, covering 847 unique conditions and 873 distinct drugs, spanning four trial phases and six recruitment statuses. Additionally, unmatched entries

were retained for transparency. These included 3963 unique unmatched conditions, 6367 rows with unmatched conditions, and 14690 rows where drug names could not be resolved. This curation produced a clean and standardised dataset for graph construction and model training.

2.2 Graph Construction and Feature Extraction

We constructed a bipartite graph in which nodes represent either drugs or diseases, and edges denote confirmed associations derived from clinical trial data. Each edge corresponds to a matched condition–drug pair, reflecting therapeutic use reported in one or more trials. To ensure biological relevance, 45 interventions identified as placebos were excluded from the graph. The resulting network consisted of 872 distinct drugs and 845 unique diseases, connected through 5026 edges. This graph provides

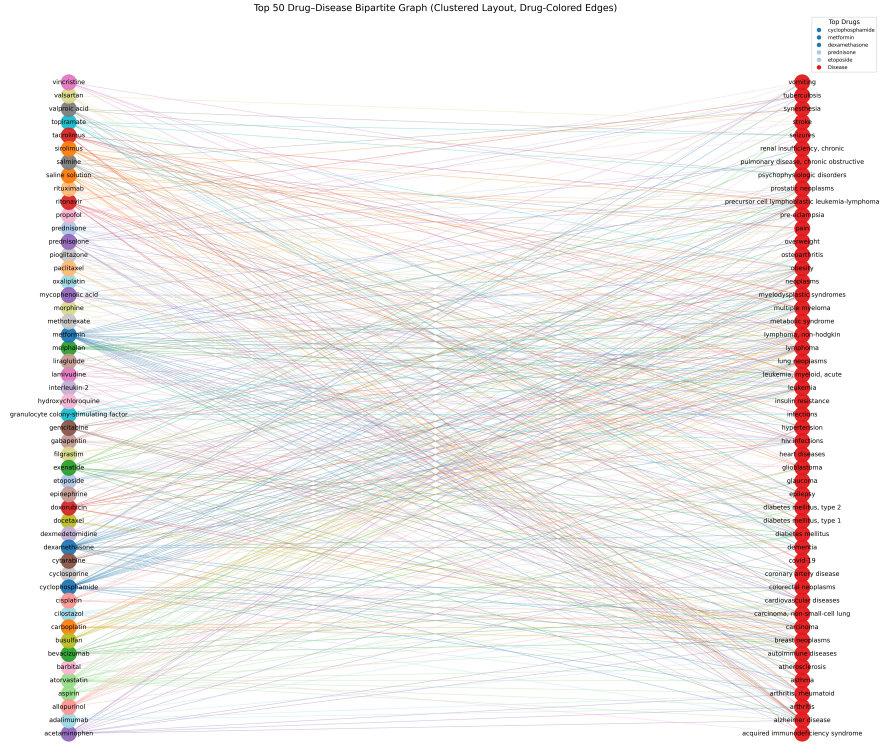


Fig. 1 Top 50 drug–disease bipartite graph. Each drug node (left) is connected to associated diseases (right) based on clinical trial matches. Edge colors correspond to individual drugs, with cluster layout highlighting frequent therapeutic overlaps.

the structural context needed to evaluate novel associations based on their position

relative to known indications. To capture this structure quantitatively, we computed five features that characterise the connectivity between each drug–disease pair:

- i Graph Distance to Known Indications quantifies the shortest number of steps between a candidate pair and any existing drug–disease association. Lower distances suggest closer proximity to established therapies.
- ii Random Walk Score estimates the likelihood that a random walker beginning at the drug node would reach the condition node. This metric reflects the influence of both direct and indirect paths within the network.
- iii Structural Likelihood models the probability of a new edge forming between the drug and disease nodes, conditioned on the existing topology. It incorporates observed patterns in how associations tend to form across the network.
- iv Preferential Attachment measures whether high-degree nodes are more likely to be connected. It captures the intuition that frequently studied drugs and prevalent conditions are more likely to co-occur.
- v Katz Similarity calculates the sum of all paths between two nodes, giving more weight to shorter connections. This metric balances local and global connectivity to highlight potential indirect therapeutic links.

2.3 Learning Feature Weights

To quantify how each topological feature contributes to the likelihood of a true drug–disease association, we trained binary classifiers on known graph edges (positives) and randomly sampled non-edges (negatives). Models included logistic regression, ridge regression, and XGBoost, selected for their ability to provide interpretable weights or feature importances. Fivefold cross-validation was used to ensure stable generalisation.

The resulting weights were normalised and used to construct the likelihood term in our Bayesian model. To evaluate their reliability, we performed robustness checks on the logistic regression weights via bootstrap resampling, Gaussian noise injection, and feature ablation. These tests confirmed the stability of weight estimates, with graph distance consistently emerging as the most informative feature. This validation helped ensure that the likelihood function remained interpretable and resistant to noise or overfitting.

2.4 Semantic Prior via Literature Classification

To embed biomedical knowledge from literature into the inference process, we developed a semantic prior derived from large language model (LLM) classification of PubMed articles. For each drug–disease pair, co-occurrence queries were submitted to PubMed Central, and we retrieved not only abstracts but also the introduction and conclusion sections of each article to enrich context. These sections were then processed in batches using a transformer-based LLM, which assigned each article to one of three categories: therapeutic (the drug helps the disease), adverse (the drug worsens or causes the disease), or irrelevant.

The semantic prior was computed by aggregating these classifications. Let T , A , and N represent the counts of therapeutic, adverse, and irrelevant articles, respectively.

The raw prior P_{raw} was defined as:

$$P_{\text{raw}} = \frac{T}{T + A + N} \quad (1)$$

To discourage overconfident associations influenced by adverse signals, we applied a penalty to the raw prior. The penalised prior $P_{\text{penalised}}$ was defined as:

$$P_{\text{penalised}} = \max\left(\frac{T - 2A}{T + A + N}, 0\right) \quad (2)$$

Next, we adjusted this prior using real-world safety data. Known side effects for each drug were fetched from the US FDA adverse event reporting system. If any reported side effects overlapped semantically with the target disease, the prior was further reduced according to the estimated severity and relevance of that overlap. Let γ represent the confidence of overlap and P_{current} the penalised prior, the enhanced prior P_{final} is computed as:

$$P_{\text{final}} = P_{\text{current}} \cdot (1 - 0.5\gamma) \quad (3)$$

This formulation reflects both literature support and potential safety risks, ensuring a more balanced representation of therapeutic potential. These final priors were normalised and passed into the Bayesian model, where they were combined with graph-based likelihoods to yield posterior predictions.

2.5 Bayesian Posterior Inference

To integrate the semantic prior and graph-based evidence into a coherent prediction, we formulated the final association score using Bayesian inference. For each drug-disease pair, we combined the prior belief (based on literature and safety data) with a likelihood function constructed from the topological features of the bipartite graph.

Let θ represent the probability that a given drug-disease pair is a true therapeutic association. The prior distribution over θ is modeled using a Beta distribution, parameterised by α and β

$$\theta \sim \text{Beta}(\alpha, \beta) \quad (4)$$

These parameters are derived from the proportion of literature classified as therapeutic (for α) and adverse or irrelevant (for β), scaled to reflect confidence and adjusted with side-effect penalties.

The likelihood of observing the graph features $\mathbf{x} = (x_1, x_2, \dots, x_5)$ given a true association is modeled using a weighted geometric mean:

$$\mathcal{L}(\mathbf{x} \mid \theta) = \prod_{i=1}^5 x_i^{w_i} \quad (5)$$

where w_i represents the normalised weight for the i^{th} feature, learned during the supervised training phase.

Using Bayes’ theorem, we compute the posterior distribution as:

$$p(\theta \mid \mathbf{x}) \propto \mathcal{L}(\mathbf{x} \mid \theta) \cdot \text{Beta}(\theta; \alpha, \beta) \quad (6)$$

This formulation updates the semantic belief with graph-derived evidence, producing a posterior that reflects both the strength of structural support and the confidence from literature. For interpretability, we summarised the posterior using two metrics: the posterior mean, which represents the expected confidence in association, and the Kullback-Leibler (KL) divergence between the prior and posterior, which quantifies the influence of topological evidence in shifting belief.

Posterior estimates were computed numerically for all candidate pairs and used to rank drug–disease associations for downstream validation and interpretation.

3 Experiment

3.1 Pipeline Walkthrough: Dexamethasone for COVID-19

To demonstrate the full pipeline of our Bayesian repurposing framework, we present a representative example evaluating dexamethasone for COVID-19.

We began by querying PubMed for articles in which "dexamethasone" and "COVID-19" co-occurred. Thirty articles were retrieved, and their abstracts, introductions, and conclusions were processed using our large language model (LLM) classifier. Out of these, 3 articles were labelled as therapeutic, 0 as adverse, and 24 as irrelevant. The raw prior probability was calculated using Equation 1, yielding $P_{\text{raw}} = 0.1111$.

Since no adverse associations were found in the literature, the penalised prior remained unchanged, i.e., $P_{\text{penalised}} = 0.1111$, as per Equation 2.

To incorporate safety evidence, we queried the FDA adverse event database for known side effects associated with dexamethasone. The following side effects were found to semantically overlap with symptoms of COVID-19: dyspnoea, cough, pneumonia, infection, fatigue, hypoxia, abdominal pain upper, dehydration, acute myocardial infarction, thrombosis in device, sepsis, respiratory failure, and death. An LLM-based adverse alignment evaluator assigned a confidence score of $\gamma = 0.85$, leading to a reduction of the prior as specified in Equation 3. The resulting enhanced prior was $P_{\text{final}} = 0.0639$.

Graph-based features were computed for the dexamethasone–COVID-19 pair using topological metrics including graph distance, random walk score, structural likelihood, preferential attachment, and Katz similarity. These were aggregated using the likelihood model described earlier, producing a structural likelihood of $\mathcal{L} = 0.9139$.

Bayesian posterior inference then integrated the enhanced prior and likelihood, yielding a posterior mean of 0.3492. The prior expectation was 0.0724, resulting in a posterior shift $\Delta\mu = 0.2768$. The KL divergence between the prior and posterior was calculated as 21.4241, reflecting a substantial update in belief supported by strong network topology. This example demonstrates how semantic priors and structural fea-

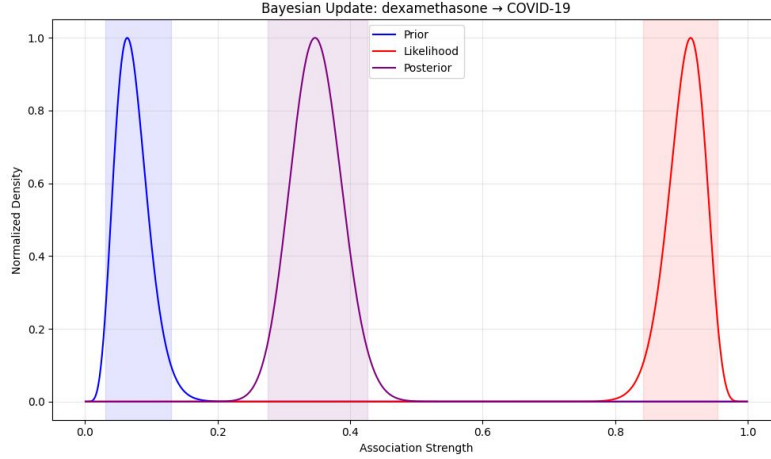


Fig. 2 Bayesian update for the dexamethasone–COVID-19 association. The prior (blue) was influenced by literature and FDA signal alignment. The likelihood (red) was derived from graph-based features. The posterior (purple) reflects the updated belief in therapeutic relevance after integrating both sources of evidence.

tures jointly inform therapeutic prediction, allowing interpretable updates even in the presence of conflicting signals such as adverse event overlap.

3.1.1 Hydroxychloroquine for Cancer-Related Indications

We applied our Bayesian repurposing framework to evaluate hydroxychloroquine across ten high-priority cancer types, selected based on global disease burden and therapeutic need. These included aggressive or hard-to-treat malignancies such as pancreatic, ovarian, and hepatocellular neoplasms.

Two known indications, breast and prostate cancer, were excluded from scoring. Among the remaining eight, we report the top five based on posterior probability and KL divergence. Table 1 presents a summary of prior and posterior expectations, structural likelihoods, and the change in belief after Bayesian updating. All values reflect the final priors after FDA adverse event correction.

Table 1 Top five candidate indications for hydroxychloroquine among selected cancer types, ranked by posterior shift and KL divergence.

Disease	KL Div.	E[Prior]	E[Likelihood]	E[Posterior]	$\Delta\mu$	Raw Prior	Final Prior
Pancreatic Neoplasms	6.80	0.3693	0.9102	0.5499	0.1806	0.3667	0.3667
Ovarian Neoplasms	12.58	0.2059	0.9105	0.4404	0.2345	0.2000	0.2000
Hepatocellular Carcinoma	19.97	0.1225	0.9803	0.4079	0.2853	0.2000	0.1150
Lung Neoplasms	15.51	0.1528	0.9095	0.4044	0.2516	0.3000	0.1458
Brain Neoplasms	16.37	0.1405	0.9113	0.3968	0.2562	0.1333	0.1333

3.1.2 Colchicine for Cardiovascular Indications

We evaluated colchicine against ten major cardiovascular diseases, ranging from coronary artery disease to pulmonary embolism. Each condition was selected based on its global prevalence and high therapeutic demand.

Table 2 reports the updated beliefs after Bayesian integration, capturing both the magnitude of the belief change and the influence of prior penalisation due to side-effect overlap.

Table 2 Top five candidate indications for colchicine among selected cardiovascular diseases.

Disease	KL Div.	E[Prior]	E[Likelihood]	E[Posterior]	$\Delta\mu$	Raw Prior	Final Prior
Atherosclerosis	13.24	0.1936	0.9118	0.4326	0.2390	0.3000	0.1875
Coronary Artery Disease	16.97	0.1324	0.9122	0.3916	0.2592	0.2667	0.1250
Cardiomyopathies	16.51	0.1324	0.8991	0.3872	0.2549	0.3333	0.1250
Heart Failure	17.70	0.1225	0.9128	0.3852	0.2627	0.2000	0.1150
Peripheral Arterial Disease	18.52	0.1119	0.9137	0.3784	0.2665	0.1667	0.1042

3.1.3 Azithromycin for Neurodegenerative Indications

To explore repurposing potential in neurology, we assessed azithromycin across ten progressive neurodegenerative disorders. These included Alzheimer’s and Parkinson’s disease, as well as rarer conditions such as multiple system atrophy and spinocerebellar ataxias.

Table 3 summarises the highest-ranking targets based on Bayesian updating. Feature gaps and sparse signals limited scoring for progressive supranuclear palsy and related disorders.

Table 3 Top five candidate indications for azithromycin among neurodegenerative diseases.

Disease	KL Div.	E[Prior]	E[Likelihood]	E[Posterior]	$\Delta\mu$	Raw Prior	Final Prior
Alzheimer Disease	16.54	0.1324	0.9001	0.3875	0.2552	0.2000	0.1250
Progressive Supranuclear Palsy	24.43	0.0425	0.9803	0.3542	0.3117	0.0333	0.0333
Amyotrophic Lateral Sclerosis	21.40	0.0752	0.9128	0.3534	0.2782	0.1333	0.0667
Huntington Disease	23.11	0.0523	0.9126	0.3380	0.2857	0.0667	0.0433
Frontotemporal Dementia	23.11	0.0523	0.9125	0.3380	0.2857	0.1333	0.0433

3.1.4 Ivermectin for Infectious Disease Targets

To assess the broader repositioning value of ivermectin beyond antiparasitic applications, we examined its performance across ten infectious disease endpoints. These included both high-prevalence global burdens (such as HIV and tuberculosis) and neglected tropical diseases like cutaneous and visceral leishmaniasis.

Table 4 Posterior shifts and belief updates for ivermectin across selected infectious diseases.

Disease	KL Div.	E[Prior]	E[Likelihood]	E[Posterior]	$\Delta\mu$	Raw Prior	Final Prior
Hepatitis B	17.20	0.1225	0.8988	0.3805	0.2580	0.2000	0.1150
Leishmaniasis, Cutaneous	15.73	0.1324	0.8768	0.3797	0.2473	0.2667	0.1250
Malaria	18.64	0.1038	0.8986	0.3678	0.2641	0.1667	0.0958
Leishmaniasis, Visceral	17.23	0.1119	0.8768	0.3660	0.2541	0.2333	0.1042
COVID-19	20.09	0.0915	0.9127	0.3644	0.2729	0.2000	0.0833

3.1.5 Metformin for Kidney-Associated Disorders

We extended our framework to evaluate the repositioning potential of metformin across nine clinically important kidney-related diseases, selected for their prevalence or unmet treatment needs. These included various forms of glomerulonephritis, chronic and acute renal conditions, and renal malignancy.

Table 5 presents the top five predictions with the most pronounced posterior shifts and KL divergence.

Table 5 Bayesian updates for metformin across selected kidney conditions.

Disease	KL Div.	E[Prior]	E[Likelihood]	E[Posterior]	$\Delta\mu$	Raw Prior	Final Prior
Glomerulonephritis, Membranous	9.47	0.2753	0.9040	0.4848	0.2095	0.4333	0.2708
Glomerulonephritis, IgA	14.08	0.1732	0.9039	0.4162	0.2430	0.3333	0.1667
Nephrotic Syndrome	18.17	0.1119	0.9036	0.3750	0.2631	0.1667	0.1042
Chronic Kidney Disease	23.50	0.0662	0.9803	0.3701	0.3039	0.2000	0.0575
Polycystic Kidney Diseases	21.49	0.0711	0.9041	0.3477	0.2767	0.1000	0.0625

4 Result and Evaluation

4.1 Concordance with Clinical Trial Evidence

We assessed the alignment between model-predicted drug-disease associations and existing clinical trial evidence. The top five disease candidates for each repurposing drug, hydroxychloroquine, colchicine, azithromycin, ivermectin, and metformin, were evaluated based on posterior shifts from our Bayesian framework. These drug and disease pairs were not included in the original training graph, ensuring that all predictions represent novel, out-of-distribution inference.

For each predicted association, we queried ClinicalTrials.gov and classified trial activity into three categories: completed, ongoing (including both recruiting and active but not recruiting), and terminated. The trial counts are summarised in Table 6.

Table 6 Clinical trial activity for top five predicted indications across five repurposing candidates. None of the evaluated drugs were part of the original indication network during model training. Trial status data were retrieved from ClinicalTrials.gov and grouped into completed, ongoing (recruiting or active), and terminated.

Drug	Predicted Disease	Completed Trials	Ongoing Trials	Terminated Trials
Hydroxychloroquine	Pancreatic Neoplasms	6	5	2
	Ovarian Neoplasms	1	1	0
	Hepatocellular Carcinoma	1	0	1
	Lung Neoplasms	3	2	7
	Brain Neoplasms	2	1	0
Colchicine	Atherosclerosis	1	0	0
	Coronary Artery Disease	2	0	0
	Cardiomyopathies	0	0	0
	Heart Failure	2	0	0
	Peripheral Arterial Disease	0	0	0
Azithromycin	Alzheimer Disease	0	0	0
	Progressive Supranuclear Palsy	0	0	0
	Amyotrophic Lateral Sclerosis	0	0	0
	Huntington Disease	0	0	0
	Frontotemporal Dementia	0	0	0
Ivermectin	Hepatitis B	0	0	0
	Leishmaniasis, Cutaneous	0	0	0
	Malaria	1	0	0
	Leishmaniasis, Visceral	0	0	0
	COVID-19	5	1	5
Metformin	Glomerulonephritis, Membranous	0	0	0
	Glomerulonephritis, IgA	0	0	0
	Nephrotic Syndrome	0	0	0
	Chronic Kidney Disease	1	1	0
	Polycystic Kidney Diseases	0	0	0

4.2 Model Robustness and Stability

We assessed the stability and reliability of the learned feature weights using three complementary analyses: bootstrap resampling, Gaussian noise injection, and single-feature ablation.

1. **Bootstrap Resampling.** We conducted 100 bootstrap iterations of logistic regression to examine variability in estimated feature weights. All five features showed extremely low standard deviations (e.g., $\text{std}_{\text{Katz}} = 0.0001$), indicating high consistency in how the model prioritised evidence. The weight assigned to GraphDistanceToIndication was particularly stable (mean = 23.32, std = 0.0064).

2. **Noise Injection.** We applied Gaussian noise ($\mu = 0, \sigma = 0.01$) to all features across 50 runs. Weight distributions remained tight, with minimal drift from original means. For example, RandomWalkScore fluctuated only within ± 0.01 , and GraphDistanceToIndication again showed strong robustness (mean = 23.38, std = 0.0040), confirming resistance to perturbations in feature inputs.
3. **Feature Ablation.** Removing each feature individually and retraining the model showed that only GraphDistanceToIndication impacted performance metrics. Its removal led to a 1.79% drop in classification accuracy ($\Delta\text{Accuracy} = 0.0179$), while all other features had zero impact on both AUC and accuracy. This underscores the feature’s critical role in the model’s decision boundary.

Table 7 summarises the observed statistics across all three stability evaluations.

Table 7 Summary of weight robustness and feature sensitivity analysis. Bootstrap and noise injection results report the standard deviation of weight estimates. Ablation results reflect change in classification accuracy and AUC when the corresponding feature is removed.

Feature	Bootstrap Std Dev	Noise Injection Std Dev	Ablation $\Delta\text{Accuracy} / \Delta\text{AUC}$
GraphDistanceToIndication	0.0064	0.0040	0.0179 / 0.0042
RandomWalkScore	0.0192	0.0056	0.0000 / 0.0000
StructuralLikelihood	0.0105	0.0033	0.0000 / 0.0000
PreferentialAttachment	0.0002	0.0042	0.0000 / 0.0000
KatzSimilarity	0.0001	0.0045	0.0000 / 0.0000

4.3 Feature Importance and Contribution

To understand how individual graph-derived features support drug–disease inference, we trained five supervised models on a binary classification task: distinguishing true associations from non-links using topological features alone. These models, logistic regression, ridge classifier, L1-penalised logistic regression, random forest, and XGBoost, were not used for prediction but to inform the weighting of features within our Bayesian likelihood formulation.

Linear models provided interpretable coefficients, while tree-based models supplied feature importances based on gain metrics. As summarised in Table 8, GraphDistanceToIndication and KatzSimilarity emerged as the most consistently informative features across all classifiers. Particularly under sparsity constraints (e.g., LogisticL1, XGBoost), these features dominated the decision space, underscoring their utility in identifying high-confidence therapeutic links.

Table 8 Feature weights across trained classifiers. Coefficients from linear models and importance scores from tree-based models reflect the relative contribution of each graph feature to classifying known drug-disease pairs.

Feature	Logistic	Ridge	LogisticL1	Random Forest	XGBoost
GraphDistanceToIndication	0.1148	0.0003	1.0535	0.4063	1.0000
RandomWalkScore	0.2470	0.0000	0.0000	0.1786	0.0000
StructuralLikelihood	-0.1154	-0.0007	0.0000	0.0003	0.0000
PreferentialAttachment	-0.0410	-0.0003	0.0000	0.0390	0.0000
KatzSimilarity	1.6515	0.1646	1.5863	0.3758	0.0000

5 Discussion

5.1 Bridging Structure and Semantics in Drug Repurposing

This work presents a Bayesian framework that combines structural graph topology with semantic priors from biomedical literature, two modalities that have long been studied independently. Traditional repurposing tools often favour one axis: either leveraging connectivity in biological networks or surfacing textual evidence from the literature. Our method unifies these views through probabilistic inference, where literature-derived priors represent initial belief and network topology shapes the likelihood of therapeutic relevance.

This fusion is not merely architectural; it is epistemological. Prior knowledge, derived from natural language in scientific discourse, encodes biomedical context and emerging therapeutic hypotheses. The graph-based likelihood then interprets that belief in the structural language of proximity, centrality, and diffusion. This duality enables a richer and more interpretable form of repurposing that explains why a drug-disease pair is plausible and how the evidence shifts our belief.

5.2 Interpretability and Robustness Through Bayesian Design

A core strength of our framework is its interpretability. Rather than returning opaque scores or static rankings, it offers dynamic posterior distributions that reflect the influence of each evidence stream. Key features such as GraphDistanceToIndication and KatzSimilarity consistently dominated model contributions across classifiers (Table 8), confirming their role in capturing biologically relevant proximity within the indication graph.

This clarity is complemented by robustness. Through bootstrap resampling, noise injection, and ablation analysis (Table 7), we show that our feature weights are stable and resilient to perturbation. This is especially critical in biomedical applications where data are sparse and noisy. Notably, GraphDistanceToIndication was the only feature whose removal caused a measurable drop in performance, highlighting its unique predictive value. Such transparency enhances trust in the model’s reasoning and provides actionable signals for downstream prioritisation.

5.3 Aligning Predictions with Clinical Trial Reality

To validate the translational relevance of our predictions, we cross-referenced the top-ranked disease candidates for five repurposing drugs against the ClinicalTrials.gov database (Table 6). The results revealed a spectrum of alignment. Predictions such as hydroxychloroquine for pancreatic and lung neoplasms or colchicine for atherosclerosis and coronary artery disease, mapped well onto existing trials, including completed studies, signalling strong clinical plausibility.

Conversely, several high-scoring predictions lacked any supporting trial activity, including azithromycin for neurodegenerative conditions and metformin for rare kidney disorders. These gaps are not necessarily false positives; rather, they spotlight neglected therapeutic areas that remain underexplored. Terminated trials, particularly for hydroxychloroquine in lung neoplasms and ivermectin in COVID-19, also illustrate the importance of incorporating negative evidence into repurposing evaluation.

5.4 Limitations and Future Directions

Despite its strengths, our framework has limitations. The semantic prior depends on literature availability and is susceptible to publication bias. Disease areas with sparse or ambiguous discourse may yield underpowered priors, while well-studied conditions may dominate due to sheer volume. Similarly, our structural graph, based on trial co-occurrence, reflects historical clinical attention rather than mechanistic or molecular relevance. This introduces the risk of reinforcing existing biases in therapeutic focus. Additionally, our model does not yet incorporate pharmacokinetics, dosage information, or patient stratification, all factors crucial for real-world efficacy. The assumption of independence between graph features, while practical for tractable inference, may overlook meaningful interactions. Lastly, our use of FDA adverse event data to penalise priors reflects a conservative stance on safety but may underrepresent off-target benefits or confounding indications.

Moving forward, we envision enriching the framework with multimodal data, including molecular interaction networks, omics profiles, and temporal clinical records. Dynamic priors that evolve with emerging literature and real-world evidence could make inference more responsive. Integrating uncertainty into downstream decision making, whether for trial design, cohort selection, or regulatory review, represents another exciting frontier.

6 Conclusion

We have introduced a novel Bayesian framework for drug repurposing that harmonises structural evidence from drug-disease networks with semantic priors mined from biomedical literature. This dual-modality approach not only improves predictive accuracy but, more critically, offers interpretable and transparent insights into the mechanisms behind each prediction. By encoding uncertainty and allowing for evidence-based updates, the model represents a meaningful step toward trustworthy, data-driven therapeutic discovery.

Our findings show that the proposed method can recover clinically supported associations while also surfacing underexplored or neglected candidates, some of which lack current trial activity but exhibit strong topological and semantic alignment. The ability to quantify belief shifts and track how different types of evidence influence final prediction positions this framework as a decision-support tool for researchers, clinicians, and regulatory bodies alike.

As biomedical data sources continue to diversify and expand, methods that can responsibly integrate heterogeneous signals will become increasingly valuable. Our framework provides a foundation for such integration, offering both scientific rigor and practical utility. Future work may extend this paradigm to incorporate patient-level data, longitudinal signals, or real-world outcomes, further strengthening the link between computational repurposing and clinical translation.

Ultimately, this study offers a principled, transparent, and extensible framework for drug repurposing that bridges biomedical semantics and structural topology. By quantifying belief shifts rather than merely predicting associations, it opens the door to a more interpretable and adaptive paradigm that aligns scientific curiosity with clinical possibility.

Supplementary Information. Supplementary materials include trial meta-data files, phase-status breakdowns, and prediction summaries for all evaluated drug–disease pairs. These materials are available alongside the main manuscript and can also be accessed through the project repository at <https://github.com/Franosei/Network-Analysis-of-Drug-Repurposing>.

Declarations

Funding: Not applicable.

Conflict of interest: The author declares no competing interests.

Ethics approval and consent to participate: Not applicable.

Consent for publication: Not applicable.

Data availability: All trial-derived data used for this study were collected from publicly accessible sources, primarily ClinicalTrials.gov. Processed datasets are available from the corresponding author on reasonable request.

Materials availability: Not applicable.

Code availability: The full source code for graph construction, feature extraction, Bayesian modelling, and evaluation is openly available at: <https://github.com/Franosei/Network-Analysis-of-Drug-Repurposing>.

Author contribution: F.O. designed the study, developed the methodology, implemented the experiments, and wrote the manuscript.

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